Exponential & Logarithmic Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 6$ in multiplying and dividing rational expressions, evaluate $f(5)$.

Final Answer: 31. Substitute $x = 5$: $f(5) = 5(5) + 6 = 31$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 6x + 7$ in solving equations with rational expressions, evaluate $f(2)$.

Final Answer: 19. Substitute $x = 2$: $f(2) = 6(2) + 7 = 19$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for solving equations with rational expressions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 19.5. The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 8$ in graphical and analytical solutions to rational equations, evaluate $f(2)$.

Final Answer: 12. Substitute $x = 2$: $f(2) = 2(2) + 8 = 12$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for graphical and analytical solutions to rational equations consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 20.5. The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the sum of all real solutions to the equation $3^x + 3^{3-x} - 10 = |3^{x-1} - 3|$? Show your complete algebraic work.

Final Answer: $\log_3\left(\frac{81}{4}\right)$. To solve the equation, express all exponential terms in terms of a single substitution variable. Notice that: $3^x = 3 \cdot 3^{x-1}$ $3^{3-x} = \frac{27}{3^x} = \frac{27}{3 \cdot 3^{x-1}} = \frac{9}{3^{x-1}}$ Let $u = 3^{x-1}$. Since exponential functions with real inputs are strictly positive, we must have $u > 0$. The equation transforms into: $3u + \frac{9}{u} - 10 = |u - 3|$ Now consider the two cases determined by the absolute value: **Case 1: $u \geq 3$** Here.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 3 · AdvancedExperimental laboratory data pairs (x, y) are transformed by computing $Y = \ln(y)$. A linear regression performed on (x, Y) yields the best-fit line $Y = -0.42x + 2.85$. Write the corresponding non-linear model for y as an exponential function of x in the form $y = ab^x$, giving a and b to three decimal places. Then predict the value of y when $x = 5$.

Step 1 (Exponentiate the Linear Equation): $\ln(y) = -0.42x + 2.85 \Rightarrow y = e^{-0.42x+2.85} = e^{2.85} \cdot e^{-0.42x} = e^{2.85} \cdot (e^{-0.42})^x$. Step 2 (Compute Parameters): $a = e^{2.85} \approx 17.28775 \approx 17.288,\ b = e^{-0.42} \approx 0.65705 \approx 0.657.$ The exponential model is $y = 17.288(0.657)^x$ (or $y = 17.288e^{-0.42x}$). Step 3 (Predict for x = 5): $y(5) = 17.288 \cdot e^{-0.42 \times 5} = 17.288 \cdot e^{-2.1} \approx 17.288 \cdot 0.122456 \approx$ **2.117**.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsA sample of a radioactive isotope has an initial mass of 250 mg and decays at a continuous rate of 4.8% per day according to $M(t) = 250e^{-0.048t}$, where t is measured in days. Determine the exact daily multiplicative decay factor b in the form $M(t) = 250b^t$, and calculate the daily percentage loss rounded to two decimal places.

Multiplicative Base: Rewrite $e^{-0.048t} = (e^{-0.048})^t$: $b = e^{-0.048} \approx 0.953134.$ The exact base is $b = e^{-0.048}$ (or approximately 0.9531). Daily Percentage Loss: The percentage decay rate per day is $1 - b$: $1 - 0.953134 = 0.046866 \Rightarrow 4.69\%.$ The daily percentage loss is 4.69% (distinguished from the continuous rate of 4.80%).

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A culture of bacteria doubles in population every 3 hours. If the initial count is 250 bacteria, write an exponential model $B(t)$ for the population after t hours, and find the population after 12 hours.

Initial population $a = 250$, doubling factor $b = 2$, time interval = 3 hours: $B(t) = 250 \cdot 2^{t/3}.$ For $t = 12$ hours, the number of doubling periods is $12/3 = 4$: $B(12) = 250 \cdot 2^4 = 250 \cdot 16 = 4000$ bacteria.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Unit 2 · Exponential and Logarithmic Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 =$ **66**. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t =$ **11**.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Solve the logarithmic equation $\log_3(x+6)+\log_3(x)=3$ and eliminate any extraneous solutions.

Combine the logarithms on the left-hand side using the product rule: $\log_3[x(x+6)]=3$. Convert to exponential form: $x(x+6)=3^3=27\Rightarrow x^2+6x-27=0$. Factor the quadratic: $(x+9)(x-3)=0\Rightarrow x=-9$ or $x=3$. Check the domain restrictions in the original equation:For $x=-9$: $\log_3(-9)$ is undefined, so $x=-9$ is an extraneous solution.For $x=3$: $\log_3(9)+\log_3(3)=2+1=3$. Valid ✓.Therefore, the unique solution is $x=3$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A radioactive substance follows the decay model $M(t)=M_0e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

If 20% decays in 15 days, then 80% remains at $t=15$: $0.80M_0=M_0e^{-15k}\Rightarrow e^{-15k}=0.80$. Take the natural logarithm of both sides: $-15k=\ln(0.80)\Rightarrow k=-\frac{\ln(0.80)}{15}=\frac{\ln(1.25)}{15}\approx 0.01488\ \text{day}^{-1}$. The half-life $t_{1/2}$ satisfies $e^{-kt_{1/2}}=0.5$: $t_{1/2}=\frac{\ln 2}{k}=\frac{15\ln 2}{-\ln(0.80)}\approx\frac{15(0.6931)}{0.2231}\approx 46.6$ days.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

[Exponential, Logarithmic & Logistic Dynamics · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

Substitute $t=8$: $g(8)=5(8)+5=\mathbf{45}$. Setting $5t+5=50$ gives $5t=45\Rightarrow t=\mathbf{9}$.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Level 2 · MediumFind all real solutions to the trigonometric equation $2\cos^2\theta+3\sin\theta-3=0$ on the interval $[0,2\pi)$.

Step 1 (Convert to Single Trig Ratio): Substitute $\cos^2\theta=1-\sin^2\theta$: $2(1-\sin^2\theta)+3\sin\theta-3=0\Rightarrow 2-2\sin^2\theta+3\sin\theta-3=0$. $-2\sin^2\theta+3\sin\theta-1=0\Rightarrow 2\sin^2\theta-3\sin\theta+1=0$. Step 2 (Factor the Quadratic in $\sin\theta$): $(2\sin\theta-1)(\sin\theta-1)=0$. Step 3 (Solve the Linear Factors): $\sin\theta=1\Rightarrow\theta=\frac{\pi}{2}$.$2\sin\theta-1=0\Rightarrow\sin\theta=\frac{1}{2}$. In $[0,2\pi)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Functions & Their Graphs**

Level 1 · FoundationsFind the exact value of $\tan\left(\frac{11\pi}{6}\right)$ and $\sec\left(\frac{11\pi}{6}\right)$ using reference angles and unit circle coordinates.

Quadrant & Reference Angle: $\frac{11\pi}{6}$ lies in Quadrant IV because $\frac{3\pi}{2}$\theta_{\text{ref}} = 2\pi - \frac{11\pi}{6} = \frac{\pi}{6}.x and \(y $\cos\left(\frac{11\pi}{6}\right) = +\cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}, \quad \sin\left(\frac{11\pi}{6}\right) = -\sin\left(\frac{\pi}{6}\right) = -\frac{1}{2}.$ Calculate Ratios: $\tan\left(\frac{11\pi}{6}\right) = \frac{-1/2}{\sqrt{3}/2} = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}.$.

16 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Matrices & Systems**

Level 3 · AdvancedFind all values of θ in $[0, 2\pi)$ for which the rose curve $r = 4\sin(3\theta)$ passes through the pole (origin), and find the maximum distance of any point on the curve from the origin.

Step 1 (Find Pole Crossings where r = 0): $4\sin(3\theta) = 0 \Rightarrow \sin(3\theta) = 0.$ For $\theta \in [0, 2\pi)$, the argument $3\theta \in [0, 6\pi)$. The sine function equals zero at integer multiples of π: $3\theta = 0, \pi, 2\pi, 3\pi, 4\pi, 5\pi.$ $\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}.$ Step 2 (Find Maximum Distance from Pole): Distance from pole is $|r| = |4\sin(3\theta)|$. The maximum value of $|\sin(3\theta)|$ is 1. Therefore, the maximum distance from t.

17 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Probability & Statistics**

Let $f(x) = \frac{x^2 - 9}{x^2 - 5x + 6}$. Identify the locations of all vertical asymptotes and removable discontinuities (holes) of $f(x)$.

Factor numerator and denominator completely: $f(x) = \frac{(x-3)(x+3)}{(x-2)(x-3)}$ Notice that $(x - 3)$ is a common factor in both numerator and denominator: Removable Discontinuity (Hole): Occurs at $x = 3$. To find the y -coordinate, evaluate the simplified fraction: $f_{\text{reduced}}(3) = \frac{3+3}{3-2} = \frac{6}{1} = 6$. The hole is at **(3, 6)**.Vertical Asymptote: The remaining uncancelled denominator factor is $x - 2 = 0$, creating a vertical asymptote at line **x = 2**.

18 **[Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratics & Complex Numbers**

A bivariate dataset (x, y) is transformed into a semi-log plot $(x, \ln y)$, yielding the linear regression model $\ln(\hat{y}) = 1.5 + 0.4x$. Express $\hat{y}$ as an exponential function of x in the form $\hat{y} = a \cdot b^x$, and find the predicted percentage growth rate per unit increase in x.

Step 1: Exponentiate both sides of the semi-log regression equation: $\hat{y} = e^{1.5 + 0.4x} = e^{1.5} \cdot (e^{0.4})^x.$ Step 2: Evaluate parameters a and b: $a = e^{1.5} \approx 4.4817, \quad b = e^{0.4} \approx 1.4918 \Rightarrow \mathbf{\hat{y} = e^{1.5}(e^{0.4})^x \approx 4.482(1.492)^x}.$ Step 3: Percentage growth rate: Since $b - 1 = e^{0.4} - 1 \approx 0.4918$, $\hat{y}$ increases by approximately **49.18%** per unit increase in x.

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19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Polynomial Functions

Solve the logarithmic equation $\log_2(x+3) + \log_2(x-3) = 4$ for all real solutions x. State any extraneous roots.

Step 1: State domain restrictions: $x+3>0 \Rightarrow x>-3$ and $x-3>0 \Rightarrow x>3$, so $x>3$. Step 2: Combine logarithms: $\log_2[(x+3)(x-3)] = 4 \Rightarrow \log_2(x^2-9) = 4$. $x^2-9 = 2^4 = 16 \Rightarrow x^2 = 25 \Rightarrow x = \pm 5$. Step 3: Eliminate extraneous solutions: $x=-5$ violates $x>3$ (producing negative arguments), so it is extraneous. The unique valid solution is $x=5$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Rational Expressions

A projectile is launched from ground level with height modelled by $h(t) = -16t^2 + 64t + 80$ feet after t seconds. Find (a) the maximum height reached, and (b) the time when it hits the ground.

1. The maximum occurs at the vertex time $t = -\frac{b}{2a}$: $t = -\frac{64}{2(-16)} = \frac{64}{32} = 2$ seconds. Substitute $t=2$ to get maximum height: $h(2) = -16(2^2) + 64(2) + 80 = -64 + 128 + 80 = 144$ feet. 2. Ground impact occurs when $h(t) = 0$: $-16t^2 + 64t + 80 = 0 \Rightarrow t^2 - 4t - 5 = 0 \Rightarrow (t-5)(t+1) = 0$. Since time must be non-negative, $t=5$ seconds.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 4$ in simplifying rational expressions, evaluate $f(2)$.

a. 10 ✓

b. 13

c. 6

d. 14

Substitute $x=2$: $f(2) = 3(2) + 4 = 10$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 5$ in adding and subtracting rational expressions, evaluate $f(2)$.

a. 13 ✓

b. 17

c. 8

d. 17

Substitute $x=2$: $f(2) = 4(2) + 5 = 13$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for simplifying rational expressions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for adding and subtracting rational expressions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for multiplying and dividing rational expressions consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?

a. $-\frac{5}{2}$ ✓

b. $-\frac{7}{2}$

c. $-\frac{13}{4}$

d. $\frac{3}{2}$

To determine $f(-1)$, we first find the constants a, b, and c: 1. **Horizontal Asymptote:** For an exponential function of the form $f(x) = a \cdot b^x + c$, as the exponential term approaches zero, the function approaches its horizontal asymptote $y = c$. Given that the asymptote is $y = -4$, we have: $c = -4$ 2. **Use the Point $(0, -1)$:** Substitute $x = 0$ and $f(0) = -1$: $f(0) = a \cdot b^0 - 4 = -1$ $a(1) - 4 = -1 \Rightarrow a = 3$ 3. **Use the Point $(2, 8)$:** Substitute $x = 2$ and $f(2) = 8$.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. **20** ✓

b. 24

c. 32

d. 44

Step 1: Determine the vertical shift parameter c. For an exponential function in the form $f(x) = a \cdot b^x + c$, the term $a \cdot b^x$ approaches 0 as $x \to -\infty$ (for $b > 1$) or as $x \to \infty$ (for 00), we find $b = 2$. Step 4: Evaluate $f(3)$. The complete function is $f(x) = 3 \cdot 2^x - 4$. Evaluating at $x = 3$:
$f(3) = 3 \cdot 2^3 - 4 = 3(8) - 4 = 24 - 4 = 20$ Misconception Traps: - Choice B (24) results from computing $a \cdot b^3 = 3 \cdot 8 = 24$ but forgetting to subt.

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28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a.

a. 2

b. 3

c. **4** ✓

d. 6

Since the graph of $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$, we can set up a system of two equations: 1. $a \cdot b^1 = 12 \Rightarrow ab = 12$ 2. $a \cdot b^3 = 108$ From equation (1), we can express a in terms of b: $a = \frac{12}{b}$. Substitute this into equation (2): $\left(\frac{12}{b}\right) \cdot b^3 = 108$ $12b^2 = 108$ $b^2 = \frac{108}{12}$ $b^2 = 9$ Since b is the base of an exponential function, $b > 0$. Thus, $b = 3$. Now, substitute the value of b back into equ.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

What is the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$?

a. 4

b. **8** ✓

c. 6

d. 16

To find the value of y when $x = 3$ for the function $f(x) = 2^{x+1}$, we substitute $x = 3$ into the function: $f(3) = 2^{3+1}$ $f(3) = 2^4$ $f(3) = 16$ Therefore, when $x = 3$, $y = 16$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the half-life formula for exponential decay?

a. $A(t) = A_0(2)^{t/h}$

b. $A(t) = A_0 e^{-t}$

c. $A(t) = A_0/2^t$

d. $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$ ✓

Applying Exponential & Logarithmic Functions principles, Standard half-life model. Therefore, the verified correct option is $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve: $e^{2x} = 5$. Express in terms of $\ln$.

a. $x = \ln k**2 + 1 - k$

b. $x = \ln \frac{k**2+1}{k}$

c. $x = \frac{\ln k**2+1}{k}$ ✓

d. $x = \frac{k}{\ln k**2+1}$

Applying Exponential & Logarithmic Functions principles, $2x = \ln 5 \Rightarrow x = \ln 5/2$. Therefore, the verified correct option is $x = \frac{\ln k**2+1}{k}$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

If compound interest is $A = P(1 + \frac{r}{n})^{nt}$, which formula gives continuous compounding?

a. $A = Pe^{rt}$ ✓

b. $A = P/e^{rt}$

c. $A = P(1 + r)^t$

d. $A = Prt$

Applying Exponential & Logarithmic Functions principles, As $n \to \infty$: $A = Pe^{rt}$. Therefore, the verified correct option is $A = Pe^{rt}$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The equation $\log_a x = b$ and $\log_b x = a$ simultaneously implies:

a. No simultaneous solution

b. $a = b$

c. $x = 0$

d. $x = a^b = b^a$ ✓

Applying Exponential & Logarithmic Functions principles, Both define x: $x = a^b$ and $x = b^a$. Therefore, the verified correct option is $x = a^b = b^a$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve: $5^x = 12$. Give exact answer.

a. $x = 12/5$

b. $x = \frac{\ln 12}{\ln 5}$ ✓

c. $x = \ln 12 - \ln 5$

d. $x = \log_{12} 5$

Applying Exponential & Logarithmic Functions principles, $x = \log_5 12 = \ln 12/\ln 5$. Therefore, the verified correct option is $x = \frac{\ln 12}{\ln 5}$.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)
Write using natural log: $\log_b x =$

a. $\frac{\ln b}{\ln x}$

b. $\ln x - \ln b$

c. $\ln x \cdot \ln b$

d. $\frac{\ln x}{\ln b}$ ✓

Applying Exponential & Logarithmic Functions principles, Change of base formula. Therefore, the verified correct option is $\frac{\ln x}{\ln b}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)
The value of e to 4 decimal places is:

a. 2.1787

b. 3.1416

c. 2.6180

d. 2.7183 ✓

Applying Exponential & Logarithmic Functions principles, $e \approx 2.71828\ldots$ Therefore, the verified correct option is 2.7183.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)
Which is the inverse of $f(x) = \log_2 x$?

a. $g(x) = \log_2 x$

b. $g(x) = 2x$

c. $g(x) = x^2$

d. $g(x) = 2^x$ ✓

Applying Exponential & Logarithmic Functions principles, Exponential base 2. Therefore, the verified correct option is $g(x) = 2^x$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Convert to exponential form: $\log_5 25 = 2$.

a. $5^{25} = 2$

b. $2^5 = 25$

c. $5^2 = 25$ ✓

d. $25^2 = 5$

Applying Exponential & Logarithmic Functions principles, Log to exponential. Therefore, the verified correct option is $5^2 = 25$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Polynomial Functions

Which is the y-intercept of $f(x) = 5 \cdot 3^x$?

a. $(1, 15)$

b. $(0, 15)$

c. $(0, 5)$ ✓

d. $(0, 3)$

Applying Exponential & Logarithmic Functions principles, $f(0) = 5 \cdot 1 = 5$. Therefore, the verified correct option is $(0, 5)$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Rational Expressions

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find A/A_0 at $t = 10$.

a. 8

b. 2

c. 4 ✓

d. 10

Applying Exponential & Logarithmic Functions principles, $2^{10/5} = 2^2 = 4$. Therefore, the verified correct option is 4.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Radicals & Rational Exponents

Which of these is equivalent to $e^{2\ln 3}$?

a. $2e^3$

b. 9 ✓

c. 6

d. $\ln 9$

Applying Exponential & Logarithmic Functions principles, $e^{\ln 9} = 9$. Therefore, the verified correct option is 9.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Sequences & Series

Find the derivative of $f(x) = x^x$.

a. $x^x \ln x$

b. x^{x+1}

c. $x \cdot x^{x-1}$

d. $x^x(1 + \ln x)$ ✓

Logarithmic differentiation: $\ln y = x \ln x \Rightarrow y'/y = 1 + \ln x \Rightarrow y' = x^x(1 + \ln x)$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Conic Sections

Find the number of solutions to $e^x = x + 2$.

a. 1

b. 2 ✓

c. Infinitely many

d. 0

Graph: e^x convex, $x + 2$ linear — they intersect at 2 points for this particular line.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Trigonometric Functions

Solve: $3^x = 2^{x+1}$. Use logarithms.

a. $x = \frac{\ln 2}{\ln 3}$

b. $x = 1$

c. $x = \frac{\ln 3}{\ln 2}$

d. $x = \frac{\ln 2}{\ln 3 - \ln 2}$ ✓

$x \ln 3 = (x+1)\ln 2 \Rightarrow x(\ln 3 - \ln 2) = \ln 2 \Rightarrow x = \frac{\ln 2}{\ln 3 - \ln 2}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Functions & Their Graphs

Show that $2^{10} = 1024 \approx 10^3$. Use this to approximate $\log_{10} 2$.

a. $\log_{10} 2 \approx 3/10 = 0.3$ ✓

b. $\log_{10} 2 = 1$

c. $\log_{10} 2 = 1/2$

d. $\log_{10} 2 = 0.5$

$2^{10} \approx 10^3 \Rightarrow 10 \log_{10} 2 \approx 3 \Rightarrow \log_{10} 2 \approx 0.3$.

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